UNIT-4: EQUATIONS OF CHANGE

Time Derivatives

- A physical quantity may vary with respect to time, as well as space. Different time derivatives are defined in order to include variation with respect to spatial coordinates.

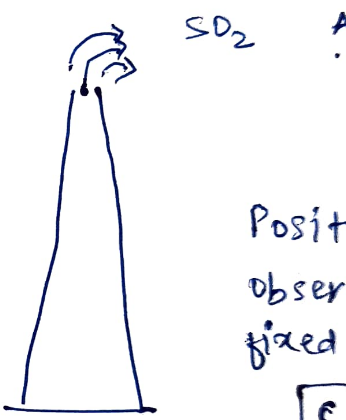
① Partial Derivative $\left(\frac{\partial c}{\partial t}\right)$

Let us consider a hypothetical case. A chimney produces flue gases. We want to measure concentration of SO_2 in the flue gas.

Let

c_1 = concentration of SO_2 at time t

$c_1 + \Delta c$ = concentration of SO_2 at time $t + \Delta t$



Position of observer is fixed at "A"

$\boxed{c = f(t)}$

At a fixed location "A", change in the concentration of SO_2 with time is

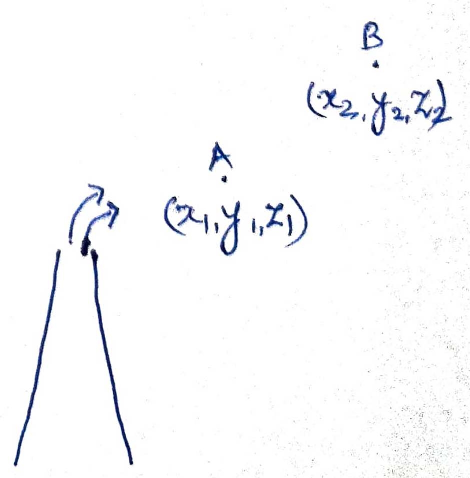
$$\lim_{\Delta t \to 0} \frac{(c_1 + \Delta c) - c_1}{\Delta t} = \lim_{\Delta t \to 0} \frac{\Delta c}{\Delta t} = \frac{\partial c}{\partial t}$$

$\boxed{\frac{\partial c}{\partial t} \text{ is measured at a fixed location}}$

② Total Derivative $\left(\frac{dc}{dt}\right)$

$\frac{dc}{dt}$ is measured when the observer keeps moving (Location is not fixed)

$c = f(x, y, z, t)$



Let

c_1 = concentration of SO_2 at time t_1 & location $A(x_1, y_1, z_1)$

c_2 = concentration of SO_2 at time t_2 & location $B(x_2, y_2, z_2)$

Time derivative is

$$\lim_{\Delta t \to 0} \frac{c_2 - c_1}{\Delta t} = \frac{dc}{dt} = \frac{\partial c}{\partial t} + v_x \frac{\partial c}{\partial x} + v_y \frac{\partial c}{\partial y} + v_z \frac{\partial c}{\partial z}$$

where, $\underline{v}$ = velocity of observer

$\underline{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$

③ Substantial derivative $\left(\frac{Dc}{Dt}\right)$

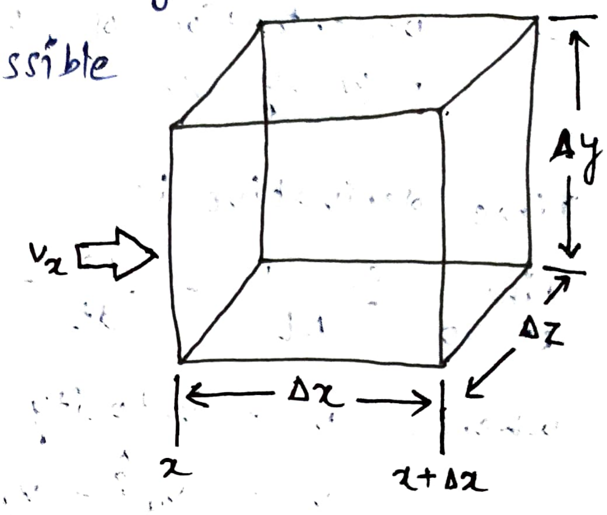
It is a special case of total derivative when the observer moves at the same velocity as the Fluid.

$$\frac{Dc}{Dt} = \frac{\partial c}{\partial t} + u_x \frac{\partial c}{\partial x} + u_y \frac{\partial c}{\partial y} + u_z \frac{\partial c}{\partial z}$$

where, $\underline{u} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k}$ is the velocity of Fluid.

(B) Equation of CONTINUITY

consider a volume element $\Delta x \Delta y \Delta z$ within flow of a compressible fluid.



Velocity of fluid

$$\vec{V} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}$$

Volumetric Flowrate of fluid entering through surface $\Delta y \Delta z$ $= \Delta y \Delta z \, V_x\big|_x$

Mass Flowrate of fluid entering through $\Delta y \Delta z$ $= \Delta y \Delta z (\rho V_x)\big|_x$

Mass Flowrate of fluid leaving through $\Delta y \Delta z$ at $x+\Delta x$

$$= \Delta y \Delta z (\rho V_x)\big|_{x+\Delta x}$$

Similarly,

Mass Flowrate IN across $\Delta x \Delta z$ $= \Delta x \Delta z (\rho V_y)\big|_y$

Mass Flowrate OUT across $\Delta x \Delta z$ $= \Delta x \Delta z (\rho V_y)\big|_{y+\Delta y}$

Mass Flowrate IN across $\Delta x \Delta y$ $= \Delta x \Delta y (\rho V_z)\big|_z$

Mass Flowrate OUT across $\Delta x \Delta y$ $= \Delta x \Delta y (\rho V_z)\big|_{z+\Delta z}$

Rate of change (or accumulation) of mass within the volume element $= \Delta x \Delta y \Delta z \dfrac{\partial \rho}{\partial t} = \dfrac{\partial m}{\partial t}$

Unsteady state Mass Balance over volume Element:

$$\begin{bmatrix} \text{Rate of mass} \\ \text{IN} \end{bmatrix} - \begin{bmatrix} \text{Rate of mass} \\ \text{OUT} \end{bmatrix} = \begin{bmatrix} \text{Rate of change} \\ \text{(or accumulation)} \\ \text{of mass} \end{bmatrix}$$

$$\Rightarrow \Delta x \Delta y \Delta z \frac{\partial \rho}{\partial t} = \Delta y \Delta z \left[(\rho V_x)\Big|_x - (\rho V_x)\Big|_{x+\Delta x} \right]$$

$$+ \Delta x \Delta z \left[(\rho V_y)\Big|_y - (\rho V_y)\Big|_{y+\Delta y} \right]$$

$$+ \Delta x \Delta y \left[(\rho V_z)\Big|_z - (\rho V_z)\Big|_{z+\Delta z} \right]$$

Divide by $\Delta x \Delta y \Delta z$

$$\Rightarrow \frac{\partial \rho}{\partial t} = -\frac{1}{\Delta x}\left[(\rho V_x)\Big|_{x+\Delta x} - (\rho V_x)\Big|_x \right] - \frac{1}{\Delta y}\left[(\rho V_y)\Big|_{y+\Delta y} - (\rho V_y)\Big|_y \right]$$

$$- \frac{1}{\Delta z}\left[(\rho V_z)\Big|_{z+\Delta z} - (\rho V_z)\Big|_z \right]$$

Taking Limit $\Delta x \to 0$, $\Delta y \to 0$, $\Delta z \to 0$.

$$\Rightarrow \frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial x}(\rho V_x) - \frac{\partial}{\partial y}(\rho V_y) - \frac{\partial}{\partial z}(\rho V_z)$$

$$\Rightarrow \boxed{\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho V_x) + \frac{\partial}{\partial y}(\rho V_y) + \frac{\partial}{\partial z}(\rho V_z) = 0} \qquad B\ 1$$

This is ~~Gn~~ General form of Eq. of continuity in Rectangular (or Cartesian) coordinate system.

Special case:

For Newtonian Fluid (or Incompressible fluid):

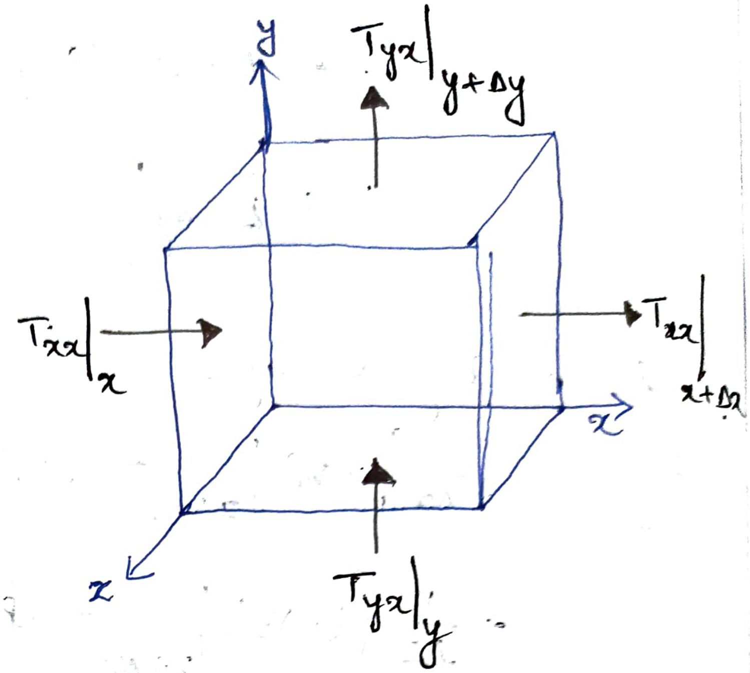
$\rho =$ constant

$$\boxed{\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = 0} \qquad B\ 2$$

(c) Equation of MOTION

Also called "Unsteady state Momentum Balance".

Let us first carry out x- momentum balance (Unsteady state) for a rectangular volume element $\Delta x \Delta y \Delta z$ within flow of a compressible fluid.



T_{ij} = Total Stress Tensor or Total Momentum Flux Tensor

Rate of x-momentum IN across surface $\Delta y \Delta z$ at x

$$= \Delta y \Delta z \, T_{xx}\big|_x$$

Rate of x-momentum OUT across surface at $x+\Delta x$

$$= \Delta y \Delta z \, T_{xx}\big|_{x+\Delta x}$$

Rate of x-momentum IN across surface at $y = \Delta x \Delta z \, T_{yx}\big|_y$

Rate of x-momentum OUT across surface at $y+\Delta y$

$$= \Delta x \Delta z \, T_{yx}\big|_{y+\Delta y}$$

Rate of x-momentum IN across surface at $z = \Delta x \Delta y \, T_{zx}\big|_z$

Rate of x-momentum OUT across surface at $z+\Delta z$

$$= \Delta x \Delta y \, T_{zx}\big|_{z+\Delta z}$$

Rate of change (or accumulation) of x-momentum within the volume element $= \frac{\partial}{\partial t}(m v_x) = \Delta x \Delta y \Delta z \frac{\partial}{\partial t}(\rho v_x)$

Body Force = $mg_x = \Delta x \Delta y \Delta z (\rho g_x)$

Unsteady State x-momentum balance

$$\begin{bmatrix}\text{Rate of} \\ x\text{-momentum} \\ \text{IN}\end{bmatrix} - \begin{bmatrix}\text{Rate of} \\ x\text{-momentum} \\ \text{OUT}\end{bmatrix} + \begin{bmatrix}\text{External} \\ \text{Forces}\end{bmatrix} = \begin{bmatrix}\text{Rate of} \\ \text{change of} \\ \text{momentum}\end{bmatrix}$$

$$\Rightarrow \Delta y \Delta z \left[\tau_{xx}\big|_x - \tau_{xx}\big|_{x+\Delta x} \right] + \Delta x \Delta z \left[\tau_{yx}\big|_y - \tau_{yx}\big|_{y+\Delta y} \right]$$

$$+ \Delta x \Delta y \left[\tau_{zx}\big|_z - \tau_{zx}\big|_{z+\Delta z} \right] + \Delta x \Delta y \Delta z (\rho g_x) =$$

$$\Delta x \Delta y \Delta z \frac{\partial}{\partial t}(\rho v_x)$$

Dividing by $\Delta x \Delta y \Delta z$ & rearranging

$$-\frac{1}{\Delta x}\left[\tau_{xx}\big|_{x+\Delta x} - \tau_{xx}\big|_x \right] - \frac{1}{\Delta y}\left[\tau_{yx}\big|_{y+\Delta y} - \tau_{yx}\big|_y \right]$$

$$-\frac{1}{\Delta z}\left[\tau_{zx}\big|_{z+\Delta z} - \tau_{zx}\big|_z \right] + \rho g_x = \frac{\partial}{\partial t}(\rho v_x)$$

Taking Limits $\Delta x \to 0$, $\Delta y \to 0$, $\Delta z \to 0$

$$\Rightarrow -\left[\frac{\partial}{\partial x}\tau_{xx} + \frac{\partial}{\partial y}\tau_{yx} + \frac{\partial}{\partial z}\tau_{zx} \right] + \rho g_x = \frac{\partial}{\partial t}(\rho v_x)$$

$$\Rightarrow \frac{\partial}{\partial t}(\rho v_x) = -\left[\frac{\partial}{\partial x}\tau_{xx} + \frac{\partial}{\partial y}\tau_{yx} + \frac{\partial}{\partial z}\tau_{zx} \right] + \rho g_x \qquad \text{(C1)}$$

Similarly, y & z-momentum balance can be written:

y-momentum

$$\frac{\partial}{\partial t}(\rho v_y) = -\left[\frac{\partial}{\partial x}\tau_{xy} + \frac{\partial}{\partial y}\tau_{yy} + \frac{\partial}{\partial z}\tau_{zy} \right] + \rho g_y$$

z-momentum

$$\frac{\partial}{\partial t}(\rho v_z) = -\left[\frac{\partial}{\partial x}\tau_{xz} + \frac{\partial}{\partial y}\tau_{yz} + \frac{\partial}{\partial z}\tau_{zz} \right] + \rho g_z$$

Total Stress Tensor $\underline{\underline{T}} = p\underline{\underline{I}} + \underline{\underline{\tau}} + \rho \underline{v}\underline{v}$

$$\Rightarrow T_{xx} = \tau_{xx} + \rho v_x^2 + p$$

$$T_{yx} = \tau_{yx} + \rho v_y v_x$$

$$T_{zx} = \tau_{zx} + \rho v_z v_x$$

substituting in Eq (C1)

$$\frac{\partial}{\partial t}(\rho v_x) = -\frac{\partial}{\partial x}(p + \tau_{xx} + \rho v_x^2) - \frac{\partial}{\partial y}(\tau_{yx} + \rho v_y v_x) - \frac{\partial}{\partial z}(\tau_{zx} + \rho v_z v_x) + \rho g_x$$

$$= -\frac{\partial p}{\partial x} - \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] - \left[\frac{\partial}{\partial x}(\rho v_x^2) + \frac{\partial}{\partial y}(\rho v_y v_x) + \frac{\partial}{\partial z}(\rho v_z v_x)\right] + \rho g_x$$

$$\Rightarrow \rho \frac{\partial}{\partial t} v_x + v_x \frac{\partial \rho}{\partial t} = -\frac{\partial p}{\partial x} - \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] - \left[v_x \frac{\partial}{\partial x}(\rho v_x) + \rho v_x \frac{\partial}{\partial x}(v_x) + v_x \frac{\partial}{\partial y}(\rho v_y) + \rho v_y \frac{\partial v_x}{\partial y} + v_x \frac{\partial}{\partial z}(\rho v_z) + \rho v_z \frac{\partial v_x}{\partial z}\right] + \rho g_x$$

$$\Rightarrow v_x\left[\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) + \frac{\partial}{\partial y}(\rho v_y) + \frac{\partial}{\partial z}(\rho v_z)\right] + \rho \frac{\partial v_x}{\partial t} + \rho v_x \frac{\partial v_x}{\partial x} + \rho v_y \frac{\partial v_x}{\partial y} + \rho v_z \frac{\partial v_x}{\partial z} = -\frac{\partial p}{\partial x} - \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] + \rho g_x$$

$$\Rightarrow \rho\left(\frac{\partial}{\partial t} + v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z}\right) v_x = -\frac{\partial p}{\partial x} - \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] + \rho g_x$$

$$\Rightarrow \boxed{\rho \frac{D v_x}{Dt} = -\frac{\partial p}{\partial x} - \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] + \rho g_x} \quad (C2)$$

where $\frac{D}{Dt} = \frac{\partial}{\partial t} + v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z}$

$\therefore$ Equations of motion are

ⓐ x-momentum

$$\rho \frac{DV_x}{Dt} = -\frac{\partial p}{\partial x} - \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] + \rho g_x$$

ⓑ y-momentum

$$\rho \frac{DV_y}{Dt} = -\frac{\partial p}{\partial y} - \left[\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z}\right] + \rho g_y$$

ⓒ z-momentum

$$\rho \frac{DV_z}{\partial t} = -\frac{\partial p}{\partial z} - \left[\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z}\right] + \rho g_z$$

Tensorial form: $\boxed{\rho \frac{D\underline{V}}{Dt} = -\underline{\nabla} p - [\underline{\nabla} \cdot \underline{\underline{\tau}}] + \rho \underline{g}}$

Special case of Eq. of motion

① Navier-Stokes Equation

For Newtonian fluids, ρ & μ are constants.

$\underline{\underline{\tau}}$ = viscous stress tensor

For Newtonian fluid,

$$\tau_{xx} = -2\mu \frac{\partial V_x}{\partial x} \qquad \tau_{yy} = -2\mu \frac{\partial V_y}{\partial y} \qquad \tau_{zz} = -2\mu \frac{\partial V_z}{\partial z}$$

$$\tau_{xy} = \tau_{yx} = -\mu\left(\frac{\partial V_x}{\partial y} + \frac{\partial V_y}{\partial x}\right)$$

$$\tau_{xz} = \tau_{zx} = -\mu\left(\frac{\partial V_x}{\partial z} + \frac{\partial V_z}{\partial x}\right)$$

$$\tau_{yz} = \tau_{zy} = -\mu\left(\frac{\partial V_y}{\partial z} + \frac{\partial V_z}{\partial y}\right)$$

substituting in (C2)

$$\rho \frac{DV_x}{Dt} = -\frac{\partial p}{\partial x} + 2\mu \frac{\partial^2 V_x}{\partial x^2} + \mu\left(\frac{\partial^2 V_x}{\partial y^2} + \frac{\partial^2 V_y}{\partial y \partial x}\right) + \mu\left(\frac{\partial^2 V_x}{\partial z^2} + \frac{\partial^2 V_z}{\partial z \partial x}\right) + \rho g_x$$

$$\Rightarrow \rho \frac{Dv_x}{Dt} = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) v_x + \rho g_x$$
$$+ \mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_y}{\partial x \partial y} + \frac{\partial^2 v_z}{\partial x \partial z}\right)$$

Last term on RHS

$$\mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_y}{\partial x \partial y} + \frac{\partial^2 v_z}{\partial x \partial z}\right) = \mu \frac{\partial}{\partial x}\left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right)$$
$$= 0 \quad (\because \text{Eq. of continuity})$$

x-momentum

$$\therefore \boxed{\rho \frac{Dv_x}{Dt} = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) v_x + \rho g_x} \quad (3)$$

y-momentum

$$\rho \frac{Dv_y}{Dt} = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) v_y + \rho g_y$$

z-momentum

$$\rho \frac{Dv_z}{Dt} = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) v_z + \rho g_z$$

Tensorial form: $\boxed{\rho \frac{D\underline{v}}{Dt} = -\underline{\nabla} p + \mu \nabla^2 \underline{v} + \rho \underline{g}}$

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special case

(2) EULER Equation

Euler equation governs the conservation of momentum for INVISCID FLOW (zero viscosity).

For $\mu = 0$, $\quad \underline{\nabla} \cdot \underline{\underline{\tau}} = 0$

Substituting in Eq. of motion

$$\boxed{\rho \frac{D\underline{v}}{Dt} = -\underline{\nabla} p + \rho \underline{g}}$$

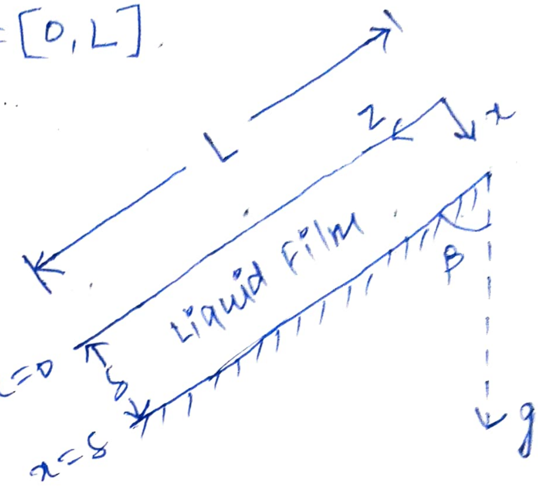
Euler's eq. is valid for fully turbulent flows when the flow becomes independent of fluid viscosity.

Use Navier-Stokes Eq. to solve steady state flow problem

Ⓐ Flow over an inclined ~~flate~~ plate

$x = [0, \delta] \quad y = [0, W] \quad z = [0, L]$.

$L \gg \delta$ and $W \gg \delta$



Assumptions:

① Steady state
② Laminar flow
③ Fully developed flow
④ Newtonian Liquid film (constant ρ & μ)

Postulates:

$v_z = v_z(x) \qquad v_x = 0 \qquad v_y = 0 \qquad p = p(x)$

Boundary conditions:

B.C. ①: At $x = 0$, $\tau_{xz} = 0 \Rightarrow -\mu \frac{dv_z}{dx} = 0 \Rightarrow \frac{dv_z}{dx} = 0$

B.C. ②: At $x = \delta$, $v_z = 0$

z-component of Navier-Stokes eq.

$$\rho \frac{Dv_z}{Dt} = -\frac{\partial p}{\partial z} + \mu \nabla^2 v_z + \rho g_z$$

$$\Rightarrow \rho\left(\frac{\partial}{\partial t} + v_x \frac{\partial}{\partial x} + v_y \frac{\partial}{\partial y} + v_z \frac{\partial}{\partial z}\right) v_z = -\frac{\partial p}{\partial z} + \mu\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) \cdot v_z + \rho g \cos\beta \quad ①$$

Steady state $\Rightarrow \frac{\partial v_z}{\partial t} = 0$

As $v_z \neq f(z) \Rightarrow \frac{\partial v_z}{\partial z} = 0$ and $\frac{\partial^2 v_z}{\partial z^2} = 0$

As $p \neq f(z) \Rightarrow \frac{\partial p}{\partial z} = 0$

As $v_z \neq f(y) \Rightarrow \frac{\partial v_z}{\partial y} = 0$ and $\frac{\partial^2 v_z}{\partial y^2} = 0$

Substituting in 1

$$\mu \frac{\partial^2 v_z}{\partial x^2} + \rho g \cos\beta = 0$$

$$\Rightarrow \frac{d^2 v_z}{dx^2} = -\frac{\rho g \cos\beta}{\mu}$$

Integrating once w.r.t. x

$$\int \frac{d^2 v_z}{dx^2} = -\frac{\rho g \cos\beta}{\mu} \int dx$$

$$\Rightarrow \frac{dv_z}{dx} = -\frac{\rho g \cos\beta}{\mu} x + c_1 \quad (2)$$

B.C. 1: At $x = 0$, $\frac{dv_z}{dx} = 0$

substituting in 2

$0 = 0 + c_1 \Rightarrow c_1 = 0$

substituting c_1 in eq. 2

$$\frac{dv_z}{dx} = -\frac{\rho g \cos\beta}{\mu} x$$

Integrating once again

$$\int dv_z = -\frac{\rho g \cos\beta}{\mu} \int x\, dx$$

$$\Rightarrow v_z = -\frac{\rho g \cos\beta}{2\mu} x^2 + c_2 \quad (3)$$

B.C. 2: At $x = \delta$, $v_z = 0$

substituting in 3

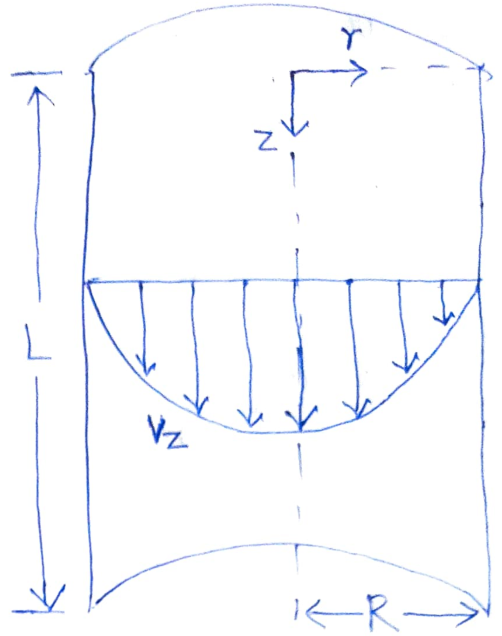
$$0 = -\frac{\rho g \cos\beta}{2\mu} \delta^2 + c_2$$

substituting c_2 in 3

$$v_z = \frac{\rho g \cos\beta}{2\mu} (\delta^2 - x^2)$$

(B) Flow Through Vertical Pipe

$r = [0, R] \quad z = [0, L] \quad \theta = [0, 2\pi]$



Assumptions:

1. Steady state
2. Laminar flow
3. Newtonian liquid (constant ρ and μ)
4. Fully developed flow.

Postulates:

$V_z = V_z(r) \qquad V_r = 0 \qquad V_\theta = 0$

$p = p(z)$

Boundary conditions:

B.C. 1: At $r = 0$, τ_{rz} is finite $\Rightarrow \frac{dV_z}{dr}$ is finite.

B.C. 2: At $r = R$, $V_z = 0$.

z-component of Navier-Stokes eq. (cylindrical coordinates)

$$\rho\left(\frac{\partial V_z}{\partial t} + V_r\frac{\partial V_z}{\partial r} + \frac{V_\theta}{r}\frac{\partial V_z}{\partial \theta} + V_z\frac{\partial V_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V_z}{\partial \theta^2} + \frac{\partial^2 V_z}{\partial z^2}\right] + \rho g_z \quad (1)$$

Steady state $\Rightarrow \frac{\partial V_z}{\partial t} = 0$

As $V_z = V_z(r) \Rightarrow \frac{\partial V_z}{\partial \theta} = 0$ and $\frac{\partial V_z}{\partial z} = 0$

Also, $\frac{\partial^2 V_z}{\partial \theta^2} = 0$ and $\frac{\partial^2 V_z}{\partial z^2} = 0$

Substituting in (1)

$$-\frac{\partial p}{\partial z} + \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V_z}{\partial r}\right) + \rho g_z = 0$$

$$\Rightarrow \frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) = \frac{\partial p}{\partial z} - \rho g$$

As p is a linear function of z

$$\frac{dp}{dz} = \frac{\Delta P}{\Delta Z} = \frac{\Delta p}{L}$$

Modified pressure can be defined as

$$\frac{\Delta P}{L} - \rho g = \frac{P_L - P_0}{L}$$

z- momentum becomes

$$\frac{\mu}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) = \frac{P_L - P_0}{L}$$

$$\Rightarrow \frac{d}{dr}\left(r\frac{dv_z}{dr}\right) = \frac{P_L - P_0}{L\mu} r$$

Integrating

$$\int d\left(r\frac{dv_z}{dr}\right) = \frac{P_L - P_0}{L\mu}\int r\,dr$$

$$\Rightarrow r\frac{dv_z}{dr} = \frac{P_L - P_0}{2L\mu} r^2 + C_1$$

$$\Rightarrow \frac{dv_z}{dr} = \frac{P_L - P_0}{2L\mu} r + \frac{C_1}{r} \quad (3)$$

B.C. (1): At $r=0$,

$$\left.\frac{dv_z}{dr}\right|_{r=0} = 0 + \frac{C_1}{0}$$

if $C_1 \neq 0$, $\left.\frac{dv_z}{dr}\right|_{r=0}$ becomes infinite.

which is not possible.

For $\left.\frac{dv_z}{dr}\right|_{r=0}$ to be finite, C_1 must be 0.

substituting in eq. (3)

$$\frac{dv_z}{dr} = \frac{P_L - P_0}{2\mu L} r.$$

Integrating w.r.t. r

$$\int dv_z = \frac{P_L - P_0}{2\mu L} \int r\,dr$$

$$\Rightarrow v_z = \frac{P_L - P_0}{4\mu L} r^2 + c_2 \quad (4)$$

B.C. (2): At $r = R$, $v_z = 0$

Substituting in (4)

$$0 = \frac{P_L - P_0}{4\mu L} R^2 + c_2 \Rightarrow c_2 = \frac{P_0 - P_L}{4\mu L} R^2$$

Substituting in eq. (4)

$$\boxed{v_z(r) = \frac{P_0 - P_L}{4\mu L}(R^2 - r^2)} \quad (5)$$

Average velocity

$$\langle v_z \rangle = \frac{\int_0^R \int_0^{2\pi} r v_z \, d\theta \, dr}{\int_0^R \int_0^{2\pi} r \, d\theta \, dr} = \frac{2}{R^2}\int_0^R r v_z(r)\,dr$$

$$= \frac{P_0 - P_L}{2\mu L R^2} \int_0^R (R^2 r - r^3)\,dr$$

$$= \frac{P_0 - P_L}{2\mu L R^2} \left[\frac{R^2 r^2}{2} - \frac{r^4}{4}\right]_0^R$$

$$= \frac{P_0 - P_L}{8\mu L} R^2$$

Volumetric flowrate

$$\dot{V} = A\langle V_z\rangle = \pi R^2 \cdot \frac{P_0 - P_L}{8\mu L} R^2$$

$$= \frac{\pi (P_0 - P_L) R^4}{8\mu L}$$

Mass Flowrate

$$\dot{m} = \rho \dot{V} = \boxed{\frac{\rho \pi (P_0 - P_L) R^4}{8\mu L} = \dot{m}}$$

This is Hagen- Poiseuille eq.

Dimensional Analysis of Equation of Motion (I)

Equations are often converted into Non-dimensional form.

→ Advantages of using Non-dimensional form of equations:

1. Makes equations independent of scalability.
2. Reduces parametric space
3. Useful Non-dimensional numbers are obtained.

Scales used for Non-Dimensionalization:

1. Characteristic Length: l_0 or R (Film thickness or Tube radius)
2. Characteristic velocity: v_0 (Average or Max. velocity)
3. Time: $\frac{l_0}{v_0}$
4. Pressure
   a. Inertial scale: ρv_0^2 (For High Re)
   b. Viscous scale: $\frac{\mu v_0}{l_0}$ (For Low Re)

Let us denote dimensional physical quantity with '*'.

1. Non-dimensional Length (x)

$$x = \frac{x^*}{l_0} \Rightarrow x^* = l_0 x$$

2. Non-dimensional velocity (v)

$$v = \frac{v^*}{v_0} \Rightarrow v^* = v_0 v^*$$

3. Non-dimensional Time (t)

$$t = \frac{v_0}{l_0} t^* \Rightarrow t^* = \frac{l_0}{v_0} t$$

4. Non-dimensional Pressure (p)

Scales used for Non-dimensionalization:

Let $x^*, v^*, t^*, p^* \longrightarrow$ ~~Non~~ dimensional Physical quantity

$x, v, t, p \longrightarrow$ Non-dimensional Physical quantity

① Length scale

Liquid film thickness or tube radius (l_0 or R)

$$x = \frac{x^*}{l_0} \Rightarrow x^* = l_0 x$$

② Velocity scale

Average or Maximum velocity (V_0)

$$v_x = \frac{v_x^*}{V_0} \Rightarrow v_x^* = V_0 v_x$$

③ Time scale $\left(\frac{l_0}{V_0}\right)$

$$t = \frac{V_0}{l_0} t^* \Rightarrow t^* = \frac{l_0}{V_0} t$$

④ Pressure

ⓐ Inertial scale (ρV_0^2) for High Re

$$p = \frac{p^*}{\rho V_0^2} \Rightarrow p^* = \rho V_0^2 p$$

ⓑ Viscous scale (for Low Re): $\frac{\mu V_0}{l_0}$

$$p = \frac{l_0}{\mu V_0} p^* \Rightarrow p^* = \frac{\mu V_0}{l_0} p$$

Eq. of continuity (Non-dimensional form)

$$\frac{\partial v_x^*}{\partial x^*} + \frac{\partial v_y^*}{\partial y^*} + \frac{\partial v_z^*}{\partial z^*} = 0$$

$$\Rightarrow \frac{V_0}{l_0}\left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right) = 0$$

$$\Rightarrow \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

Navier-Stokes Eq. (Non-dimensional form)

x-momentum

$$\rho\left(\frac{\partial}{\partial t^*} + v_x^*\frac{\partial}{\partial x^*} + v_y^*\frac{\partial}{\partial y^*} + v_z^*\frac{\partial}{\partial z^*}\right)v_x^* = -\frac{\partial p^*}{\partial x^*} + \mu\left(\frac{\partial^2}{\partial x^{2*}} + \frac{\partial^2}{\partial y^{2*}} + \frac{\partial^2}{\partial z^{2*}}\right)v_x^* + \rho g_x \quad \text{(D1)}$$

ⓐ Using Inertial scale

$$\Rightarrow \frac{\rho V_0}{l_0}\left(\frac{\partial}{\partial t} + v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z}\right)(V_0 v_x) = \frac{\rho V_0^2}{l_0}\left(-\frac{\partial p}{\partial x}\right) + \frac{\mu V_0}{l_0^2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)v_x + \rho g_x$$

Divide by $\frac{\rho V_0^2}{l_0}$

$$\Rightarrow \left(\frac{\partial}{\partial t} + v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z}\right)v_x = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 v_x + \frac{l_0}{V_0^2}g_x \quad \text{(D2)}$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ & $Re = \frac{\rho V_0 l_0}{\mu}$

ⓑ Using viscous scale

Eq. (D1) can be written as

$$\frac{\rho V_0^2}{l_0}\left(\frac{\partial}{\partial t} + v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z}\right)v_x = -\frac{\mu V_0}{l_0^2}\frac{\partial p}{\partial x} + \frac{\mu V_0}{l_0^2}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)v_x + \rho g_x$$

Divide by $\frac{\mu V_0}{l_0^2}$

$$Re\left(\frac{\partial}{\partial t} + v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z}\right)v_x = -\frac{\partial p}{\partial x} + \nabla^2 v_x + \frac{\rho g_x l_0^2}{\mu V_0} \quad \text{(D3)}$$

Eq. (D2) & (D3) can be written as

ⓐ High Re

$$\boxed{\frac{Dv_x}{Dt} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\nabla^2 v_x + 1/Fr}$$

Froude number, $Fr = \frac{V_0^2}{l_0 g_x}$

ⓑ Low Re

$$\boxed{Re\frac{Dv_x}{Dt} = -\frac{\partial p}{\partial x} + \nabla^2 v_x + \frac{Re}{Fr}}$$

$$\frac{Re}{Fr} = \frac{l_0 V_0 \rho}{\mu}\frac{l_0 g_x}{V_0^2} = \frac{\rho l_0^2 g_x}{\mu V_0}$$